

Vision-Based Terrain Classification for Robotic Navigation Using Classical Machine Learning

Project Midterm Report

Author: Aaron

Abstract

This project investigates terrain classification for robotic navigation using classical computer vision and machine learning techniques. The goal is to classify terrain types such as grass, gravel, and asphalt from image patches extracted from the RUGD dataset. Rather than relying on deep learning, this work focuses on interpretable feature extraction using Histogram of Oriented Gradients (HOG), which captures local edge and texture information. A custom HOG implementation was developed and validated against a standard library implementation, achieving strong agreement. These features are used to train Support Vector Machine (SVM) classifiers with different kernels. Preliminary results demonstrate that classical texture-based methods can achieve promising classification performance, with accuracy exceeding 90% on some subsets. This work establishes a baseline for efficient terrain classification and provides a foundation for future improvements including dimensionality reduction and additional feature descriptors.

1 Introduction

Terrain classification is an important perception task in autonomous robotics. Mobile robots operating in outdoor environments must be able to distinguish between different surface types in order to navigate safely and efficiently. Terrain properties such as roughness, texture, and material composition directly affect traction, stability, and path planning decisions.

This project focuses on classifying terrain types from visual input using classical machine learning techniques. The dataset used is the RUGD (Robotics Urban Ground Dataset), which provides RGB images along with pixel-level annotations of terrain classes such as grass, gravel, dirt, and asphalt. To transform this segmentation dataset into a classification problem, images are divided into fixed-size patches, and each patch is assigned a label based on the dominant terrain class within it.

Rather than using raw pixel values, this work extracts structured features that capture texture and gradient information. The primary feature used in this stage is Histogram of Oriented Gradients (HOG), which encodes distributions of edge orientations across local regions of the image. These features are then used to train Support Vector Machine (SVM) classifiers.

The expected outcome is a model capable of distinguishing terrain types with reasonable accuracy while remaining computationally efficient. This approach also serves as a strong baseline for comparison with more complex methods.

2 Related Work

Terrain classification has been widely studied in both robotics and computer vision. Many modern approaches rely on deep learning, particularly convolutional neural networks (CNNs), which can learn hierarchical features directly from image data. However, these methods often require large datasets and significant computational resources.

Prior to the rise of deep learning, classical feature descriptors such as Histogram of Oriented Gradients (HOG) and Local Binary Patterns (LBP) were commonly used for image classification tasks. HOG, introduced by Dalal and Triggs, is particularly effective at capturing edge and shape information, while LBP focuses on local texture patterns. These features have been successfully applied in object detection and texture classification tasks.

Support Vector Machines (SVMs) are also a well-established method for classification in high-dimensional feature spaces. They are particularly effective when combined with engineered features such as HOG, as they can construct decision boundaries that generalize well even with limited data.

This project builds on these classical methods, applying them to terrain classification in a robotics context and evaluating their effectiveness relative to different kernel choices.

3 Histogram of Gradients and Feature Extraction

This section describes the data processing pipeline and the implementation of Histogram of Oriented Gradients (HOG), which forms the core feature representation used in this project.

3.1 Image Pre-processing

Significant effort was devoted to preparing the dataset for feature extraction. Each RGB image is converted to grayscale to reduce dimensionality while preserving structural information. This is appropriate since HOG primarily depends on intensity gradients rather than color.

Pixel intensities are normalized to the range $[0, 1]$ to improve numerical stability. Additionally, logarithmic gamma normalization is used to improve lighting invariability in the HOG extraction.

Images are divided into 64×64 patches. These patches are labeled by computing the distribution of terrain classes within the corresponding segmentation map and selecting the dominant class. To reduce label noise, patches with highly mixed terrain composition can be filtered out.

An efficient tiling strategy was also implemented to handle images whose dimensions are not exact multiples of the patch size. This minimizes information loss due to discarded border regions. This was ultimately not used in training as it introduces image size inconsistency to the HOG extraction.

Importantly, the dataset is split at the image level (rather than patch level) into training and testing sets. This prevents patches from the same image from appearing in both sets and, which would otherwise lead to overly optimistic performance estimates as the model would not be required to generalize for unseen lighting and camera orientations.

3.2 HOG Feature Extraction

A custom implementation of Histogram of Oriented Gradients was developed using NumPy. The algorithm proceeds as follows:

- Image gradients are computed using 3×3 Sobel filters, approximating partial derivatives in the x and y directions.
- Gradient magnitude and orientation are computed at each pixel.
- Pixel-level computations use symmetric padding about the image border to extend the continuous gradients and prevent an artificial cell-edges.
- The image is divided into 8×8 pixel cells, forming an 8×8 grid for each patch.

- For each cell, a histogram of 9 orientation bins is constructed over the range $[0, 180)$ degrees.
- Gradient magnitudes are distributed between adjacent bins using linear interpolation to reduce quantization artifacts.
- Cells are grouped into overlapping 2×2 blocks, and each block is normalized using L2-Hys normalization to improve robustness to illumination and contrast changes (prevent dominance of hard edges).

Each block produces a 36-dimensional feature vector ($4 \text{ cells} \times 9 \text{ bins}$). These block features are concatenated to form the final feature vector for each image patch, resulting in a high-dimensional representation.

3.3 Model Training

The extracted HOG features are used to train Support Vector Machine classifiers using the scikit-learn library. Both linear and nonlinear kernels (polynomial and RBF) are evaluated. These models are well-suited for high-dimensional feature spaces and provide a strong baseline for classification performance.

4 Preliminary Results

Preliminary experiments were conducted using subsets of the dataset due to computational constraints. The results demonstrate that HOG features combined with SVM classifiers can successfully distinguish between terrain types.

HOG comparison with scikit-image

Validation of the HOG implementation showed strong agreement with the scikit-image implementation.

- Correlation = 0.953
- Relative difference = 0.21 $[\text{L2-norm}(\text{diff}) / \text{L2-norm}(\text{scikit-image})]$

This confirms high-level correctness of the feature extraction pipeline.

Model evaluation

The following results were obtained on from training on two subsets of the dataset:

Trail9 Dataset

- Linear SVM: Accuracy = 0.8685
- Polynomial SVM: Accuracy = 0.9152
- RBF SVM: Accuracy = 0.8496

Village Dataset

- Linear SVM: Accuracy = 0.7562
- Polynomial SVM: Accuracy = 0.7733
- RBF SVM: Accuracy = 0.7952

These results suggest that nonlinear kernels may better capture the complexity of terrain distributions, though performance varies across datasets. It is important to note that these experiments were conducted on limited subsets with specific terrain compositions (grass and gravel), and broader evaluation is needed.

5 Future plan

Future work will focus on improving both the feature representation and the evaluation methodology. Planned steps include:

- Applying dimensionality reduction techniques such as Principal Component Analysis (PCA) to reduce feature dimensionality and improve efficiency.
- Expanding experiments to the full dataset with more diverse terrain types.
- Evaluating performance on less filtered patches to better reflect real-world conditions.
- Implementing Local Binary Patterns (LBP) and combining them with HOG features.
- Improving computational efficiency of the HOG implementation.
- Visualizing predictions by reconstructing terrain maps from patch-level classifications.
- Generalizing the HOG pipeline to handle arbitrary image sizes.
- Exploring more realistic train/test splits across different environments to better assess generalization.

These improvements aim to produce a more robust and realistic terrain classification system.

6 References

RUGD Dataset: <http://rugd.vision>

Dalal, N., and Triggs, B. “Histograms of Oriented Gradients for Human Detection.” IEEE CVPR, 2005.

Pedregosa et al. “Scikit-learn: Machine Learning in Python.” Journal of Machine Learning Research, 2011.